The Misguided Debate Over the Existential Import of Aristotle's Categorical Forms

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1 Introduction

The history of logic can be broadly divided into classical, Aristotelian logic, and contemporary, Fregean logic. The dissonance between these two approaches has led to many debates, one of which concerns the existential import of Aristotle's traditional categorical forms (i.e., A, E, I, O). I examine the two orthodox positions in this debate: that affirmative expression forms have existential import while negative expression forms do not (the Classical View), and that particular expression forms have existential import while universal expression forms do not (the Contemporary View). I argue that neither of these approaches is preferable, as both make the assumption that there can be a uniquely accurate, general method for assigning a propositional form to ordinary language expressions of the categorical form A. I further argue that we ought not discard the surrounding conclusions of either approach; both can be preserved as useful logical tools.

2 Terminology

The disagreement about the existential import of the categorical forms is largely caused by ambiguity. To remedy this, I define my terms. I consider *facts* to be ways in which objects are related to one another; they are true propositions. *Propositions*, then, are possible facts: ways in which possible objects could be related to one another; they are the meanings of propositional signs. I represent propositions in Quantificational Logic (QL).[1] *Propositional signs*, then, are expressions which represent a proposition. *Expressions* are collections of terms in ordinary language (OL), meaning conversational language, which are spoken, written, or otherwise manifest (e.g., grooves in a record player). We can call expressions that are not propositional signs *nonsense* (e.g., 'Purple is equal').[2] I do not claim these are *correct* or *best*; they serve only to clarify the present discussion. The debate can now be stated precisely:

> An *expression* has existential import iff it corresponds to a proposition that states the existence of the expression's subject (S). That is, a proposition expressible in QL containing (a non-negated instance of) $\exists x[Sx]$.

Broadly speaking, logic is concerned with propositions. OL expressions that are non-propositional (nonsense by our definition, e.g., 'Hello,' 'Open the door,' &c.) fall under the philosophy of language: logic does not directly deal with them, as they have no corresponding propositions. Hereinafter, all relevant propositions can be written in QL; expressions not representable in QL will be considered nonsense.[3] 'Tá duine ard amháin ar a laghad ann' and 'There is at least one tall person' both represent, in OL, the same proposition: $\exists x[Px \wedge Tx]$.

[1] Of course, these QL expressions are themselves propositional signs, not propositions. Nevertheless, this notational difference enables us to clearly distinguish talk about propositions from talk about propositional signs.

[2] One may further divide this category into 'ill-formed' expressions and 'ambiguous' expressions (which may represent some proposition *in context*). However, as discussed in §4, both sides make general—that is, contextless—claims, each supposing a superior, general method for interpreting all expressions of the forms A, E, I and O. The ill-formed/ambiguous distinction, then, is irrelevant here. There is no context with which one may disambiguate ambiguous general expression forms.

[3] Not all meaningful expressions are expressible in QL. I claim only that those relevant here can be classified either as 'nonsense' or 'parsable in QL'. Meaningful expressions that are not parsable in QL—such as modal expressions, self-referential expressions, &c.—elude the scope of this essay.

3 Two Views of Existential Import

Aristotle's traditional 'Square of Opposition' comprises four general expression forms and their relations to one another. They are traditionally labelled as follows, where S and P are general placeholders for the subject and predicate terms:

A Every S is P

E No S is P

I Some S is P

O Not every S is P

Their relations are typically represented as in Figure 1. Although often referred to as 'propositional forms,' they are schemata for expressions. As such, I refer to them as 'expression forms' throughout.

The two primary approaches to the existential import of these forms are as follows:

The Classical View The affirmative expression forms (A, I) have existential import and the negative forms (E, O) do not. This view is typically associated with Aristotle and nominalists after Ockham.[4]

The Contemporary View The particular expression forms (I, O) have existential import and the universal forms (A, E) do not.[5]

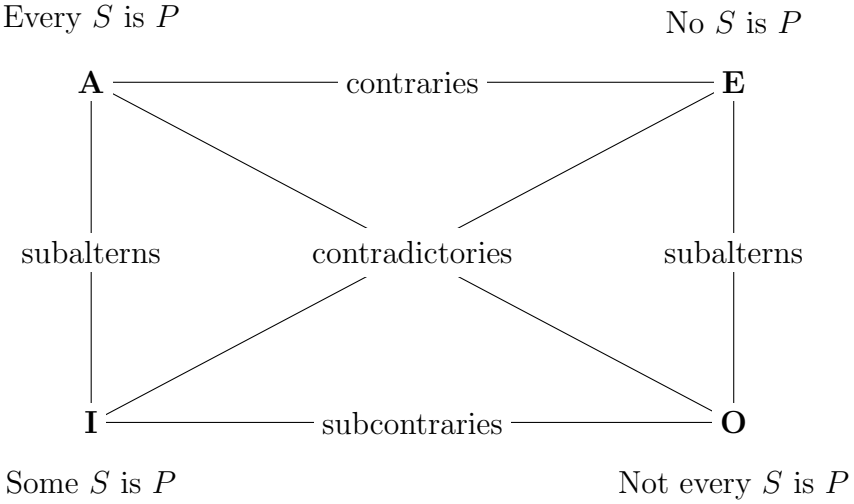


Figure 1: Traditional Square of Opposition

[4]Klima 2009, p. 144; Aristotle 1981, 1011b25; 1938, 13b12.

[5]Russell 2009, p. 62; Frege 1879, §12, pp. 134–135. See also Quine (1948).

Both approaches agree that E has no existential import and that I has existential import, but differ over A and O. I is both affirmative and particular, and E is both negative and universal. As such, I meets both conditions for existential import, and E meets neither. When considering logical structure, propositions are distinguished by truth conditions. That is, $(\sim P) \vee Q$ is equivalent to $P \rightarrow Q$ because they share the same truth conditions, even though the expression '$(\sim P) \vee Q$' is not equivalent to '$P \rightarrow Q$'. Accordingly, the disputed forms, A and O, are of particular interest here.

I favour Ackrill's translation of the expression form O, 'Not every S is P,' rather than 'Some S is not P'.[6] In the context of Aristotle's work, this seems most appropriate because it highlights that O is the contradictory of A: $Axy = (\sim Oxy)$ for all x and all y.[7] Understanding O simply as $\sim A$, and vice versa, narrows the disagreement between the Classical and Contemporary approaches to the expression form A alone (or, if one prefers, O alone). One's conclusion regarding the existential import of A immediately entails her conclusion regarding $\sim A$ (O). As such, focusing our efforts on A suffices to address the debate as a whole.

The approaches differ in practice only in the event that S does not refer: if S *does* refer, then $\exists x[Sx] \wedge \forall x[Sx \rightarrow Px]$ and $\forall x[Sx \rightarrow Px]$ must have the same truth value. Every differing truth value between the two approaches stems from their different approaches to this particular circumstance. Comparing the two approaches to form A when S does not refer, then, effectively compares their treatment of the existential import of the categorical forms as a whole.

Considering this especially relevant case—wherein S does not refer in an expression of the form A—both approaches assign specific propositional forms to the same expression form, namely, A. The Contemporary View (particular-universal approach) assigns $\forall x[Sx \rightarrow Px]$, and the Classical View (affirmative-negative approach) assigns $\exists x[Sx] \wedge \forall x[Sx \rightarrow Px]$. I will argue, however, that assigning any such propositional form to this OL expression form is arbitrary and ought to be avoided. It is in assigning any general propositional form to an expression of the form 'Every S is P' that *both* approaches are mistaken.

[6] Aristotle 1963; Parsons & Ciola 2025, §2.2.

[7] See Keynes (1906, p. 145). The contradictory relations remain on either approach to existential import (see §5). This translation choice therefore has no bearing on the debate; it does not prefer one interpretation over the other.

4 The Argument from Ambiguity

The existence of the very debate with which this essay is concerned indicates the ambiguity of expressions of the form A. 'Every S is P' is ambiguous insofar as two fluent speakers may understand it differently as to whether or not it states that there exist things that are S. If we are to properly interpret an ambiguous expression in OL, then we must understand what is meant by the speaker.[8] Consider the following example: Seán owns a pet chicken who has recently eaten her portion of seeds and will not eat again for some time. Simultaneously, Seán is cooking a chicken breast for lunch, and it has just reached the point where it is safe to eat. We cannot assign a truth value to the expression 'Seán's chicken is ready to eat' without first clarifying what it means; that is, what proposition it corresponds to: *Edible(Chicken-Poultry)* or *Hungry(Chicken-Pet)*. Without disambiguating the meaning of the expression through external context, we cannot decide how to interpret it. In OL, finding out what is meant by an expression is done intuitively, based on the surrounding context. However, in isolated or abstract cases—such as the categorical forms with which we are dealing—there is no such context. Thus, we have no basis for rearticulating expressions of form A such that they are propositional signs. The form A is therefore ambiguous, not a propositional sign; hence, by our definition, strictly 'nonsense'.

In logic, expressions are useful only insofar as they represent propositions; if an expression is not a propositional sign, then it is beyond logic's purview. Of course, we may still draw conclusions about the two propositional forms that the two approaches assign to the expression form A, but we can do so only by treating these propositional forms as independent starting points. We cannot say anything in logic about the ambiguous expression of form A in and of itself, as it corresponds to no particular proposition. Thus, under our definitions, any expression of form A is strictly nonsense. This highlights the mistaken assumption shared by both approaches to existential import, namely: 'There can be a standard method of determining the existential import (directly or indirectly) of expressions of the form A that is correct and superior to other potential methods'. Just as with 'Seán's chicken is ready to eat,' 'Every S is P' is ambiguous, and thus nonsense; we cannot make use of such expressions in logic without first determining their intended meaning.

[8] To 'properly interpret' an OL expression is to rearticulate it as a propositional sign of the proposition that the expression's originator intended to represent.

5 Preserving the Debate's Utility

The field of logic should acknowledge the mistaken assumption leading to this debate, but it need not disregard the fruits of the debate. Instead, we may draw conclusions about the propositional forms: A^{P1}: $\exists x[Sx] \wedge \forall x[Sx \rightarrow Px]$, and A^{P2}: $\forall x[Sx \rightarrow Px]$. These are distinct propositional forms, both of which are associated with the expression form A: A^{P1} on the Classical View and A^{P2} on the Contemporary View. Neither A^{P1} nor A^{P2} necessarily corresponds to A, and they can be restated more clearly in OL as 'There is at least one thing which is S, and if something is S then it is P' (S^1) and 'If something is S then it is P' (S^2) respectively. An expression is 'assigned a truth value' only in the sense that it corresponds to a proposition with a truth value; e.g., 'some men are hungry' has a truth value insofar as it represents the proposition $\exists x[Mx \wedge Hx]$. Recognising that expressions of the form A are strictly nonsense, then, prohibits asking 'Is "Every S is P" true?' (for any given S and P). Rather, we should seek a clarified propositional sign as a starting point (e.g., S^1 or S^2).[9] This leaves us with two coexisting (and indeed useful) 'squares of opposition,' as follows:

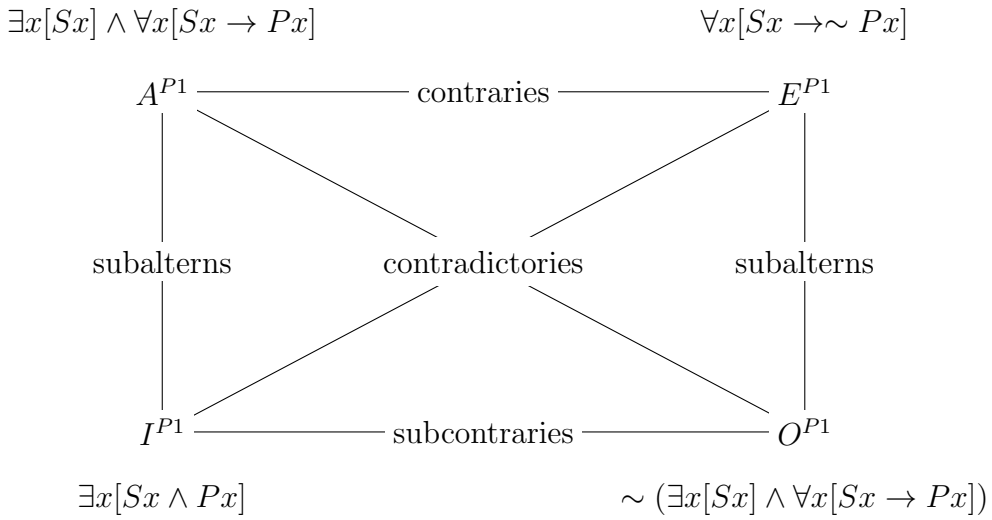


Figure 2: The Square of Opposition Regarding A^{P1}

The square in Figure 2 takes A^{P1}, rather than the expression form A, as its starting point. All that Aristotle attributed to the form A—its logical relations, laws of inference, and conclusions—in fact applies instead to A^{P1} (insofar as they were well-reasoned conclusions to begin with), as this was the meaning with which he

[9] An expression of the form A is ambiguous even if S refers, but in this case A^{P1} and A^{P2} have the same truth value. A can be clarified in this case, but doing so would not be particularly impactful. Similarly, 'Michael Collins is dead' is true, whether we refer to the former astronaut or the former Irish minister for finance.

was concerned. For example, we can replace Aristotle's conclusion that A validly obverts to E with the conclusion that A^{P1} validly obverts to $\forall x[Sx \rightarrow\sim (\sim Px)]$.[10] Unfortunately, these conclusions were tainted by starting from the false assumption that A necessarily means A^{P1}. Therefore, the logical system that lends itself to the Classical View is still useful; we just require the starting point A^{P1}, rather than A.

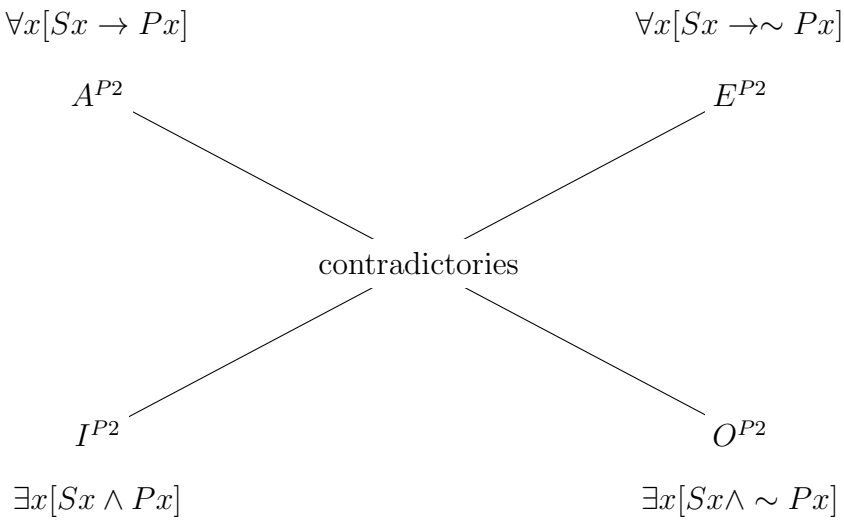


Figure 3: The Square of Opposition Regarding A^{P2}

As in the previous diagram, all the conclusions Russell and contemporary logicians draw about A (insofar as they are well reasoned to begin with) remain valid, but instead of pertaining to A, they pertain to A^{P2}, and say nothing general about expressions of form A directly. In the case of A^{P2}, only the contradictory relations hold true.

6 Conclusion

The debate over which interpretation of the existential import of A is 'correct,' or even which is preferable, is misguided; an expression of form A is ambiguous. We ought to focus on what conclusions we can draw from propositional signs and their corresponding propositions, and typically try to avoid expressions of form A, replacing them with expressions of form S^1 or S^2 according to our intended meaning. Neither way of interpreting A is generally preferable. Thus, neither interpretation of O—the contradictory of A—is preferable. Thus, neither approach to existential import of the categorical forms is preferable. Despite this, conclusions that have followed from a given understanding of A are still useful for drawing conclusions

[10]Cf. Keynes 1906, p. 101.

about propositions of the form that they (mistakenly) generally assign to A. Going forward, we should not approach the existential import of the traditional forms in either of the popularly proposed ways, but neither should we discard their teachings. Both sides of this misguided debate are useful, albeit concerned with different propositional forms or meanings.

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